Defining pyromes and global syndromes of fire regimes

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Fire is a ubiquitous component of the Earth system that is poorly understood. To date, a global-scale understanding of fire is largely limited to the annual extent of burning as detected by satellites. This is problematic because fire is multidimensional, and focus on a single metric belies its complexity and importance within the Earth system. To address this, we identified five key characteristics of fire regimes—size, frequency, intensity, season, and extent—and combined new and existing global datasets to represent each. We assessed how these global fire regime characteristics are related to patterns of climate, vegetation (biomes), and human activity. Cross-correlations demonstrate that only certain combinations of fire characteristics are possible, reflecting fundamental constraints in the types of fire regimes that can exist. A Bayesian clustering algorithm identified five global syndromes of fire regimes, or pyromes. Four pyromes represent distinctions between crown, litter, and grass-fueled fires, and the relationship of these to biomes and climate are not deterministic. Pyromes were partially discriminated on the basis of available moisture and rainfall seasonality. Human impacts also affected pyromes and are globally apparent as the driver of a fifth and unique pyrome that represents human-engineered modifications to fire characteristics. Differing biomes and climates may be represented within the same pyrome, implying that pathways of change in future fire regimes in response to changes in climate and human activity may be difficult to predict.

fire-climate-vegetation feedbacks | energetic constraints | fire intensity | fire return period | fire size

Fires occur with varying regularity and severity across almost every biome on Earth. Like vegetation, the fire that occurs at a point in space is controlled by environmental characteristics, and should change in a predictable manner along environmental gradients. In contrast to vegetation, a definition of global-scale units of fire is lacking.

Fire is often described in terms of a fire regime, which represents a particular combination of fire characteristics, such as frequency, intensity, size, season, type, and extent (1, 2). It describes the repeated pattern of fire at a location in space. Fire regimes were originally used to explain plant responses to fire (such as resprouting or seroteny) (1). However, characterizing fire regimes is also necessary when quantifying emissions from fires (3) and planning fire suppression and control (4), and is particularly important if we hope to predict how patterns of fire might change in response to environmental and human drivers (5, 6). At global scales, fire regimes could be seen as analogous to biomes.

To date, global analyses have used satellite-derived active fire and burned area data to describe the extent, interannual variability, and seasonality of burning (3, 7–9). This has contributed to an emerging global theory of fire that highlights two major energetic controls of burned area: fuel and weather (10–13). Fuel and weather show opposite trends along a productivity gradient: low-productivity environments usually do not produce enough fuel, whereas in high-biomass environments the fuel is usually too wet to burn.

We expect that such energetic considerations should act to constrain other fire characteristics to a greater or lesser extent. For example, fire intensity is strongly determined by the amount of fuel available, whereas fire frequency and fire seasonality might be more closely related to the probability of flammable conditions.

At global scale we also expect to find tradeoffs between fire characteristics, not unlike the ecological tradeoffs identified between plant traits (14) whereby not all combinations are equally probable or even possible. Where fire is frequent (i.e., every 1–3 y), fuel loads capable of producing high-intensity fires would not have enough time to accumulate. Hence, fire frequency and maximum fireline intensity should be negatively related, and we would not expect a fire regime with frequent, high-intensity fires to exist.

Some of the most pressing fire science and management questions concern changes in fire regimes caused by the introduction of invasive species (15), altered ignition patterns (16), or climate change (17). To determine how permanent these changes are and what sorts of fire regimes are emerging in these ecosystems, we need better predictive understanding of relationships with climate, vegetation, and human drivers.

These relationships are difficult to define statistically as a result of feedbacks within the fire–vegetation–climate system. A fire that occurs at a point in space is a product of the vegetation and climate; vegetation is strongly controlled by climate and fire (18, 19); and (over longer time scales) vegetation and fire in turn affect climate through altering fluxes of energy, water, carbon, and the optical and radiative properties of the atmosphere (20, 21). Correlative studies are unsatisfactory because they have to flatten the system by representing one of the elements as a response and the others as drivers.

Process-based models such as coupled climate–vegetation models can represent much of the complexity of the fire–vegetation–climate system. However, their very complexity prevents useful generalization. Plant community ecology and functional ecology (18, 22) have proven useful for understanding global patterns of vegetation. We draw on this rich theory in an analysis of global fire characteristics and their links to vegetation and climate.

Conceptually, global vegetation units (i.e., biomes) are characterized solely by the traits of vegetation (18), and the environmental correlates emerge post hoc and help to explain the boundaries between biomes. Similarly, we identify five key fire regime characteristics that can consistently be quantified at global scales by using remotely sensed imagery. We determine the leading dimensions of variation in fire, and assess which combinations of fire characteristics are more probable. By using clustering analyses, we group regions with similar fire characteristics and identify global units of fire (i.e., pyromes).

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This approach ensures that we do not confound our response variable because we classify pyromes independently of vegetation and climate, allowing them to emerge from the multidimensional space occupied by the fire characteristics. Post hoc, we assess how closely associated biomes and pyromes are, and which climate and human variables drive variation in global fire regimes. The circumpolar boreal forest biome extends through North America and Russia, but fire regimes on the two continents differ: one is characterized by crown fires, and the other by surface fires (23). This suggests that a pyrome map of the world might look somewhat different from a biome map. In contrast, the probability of fire and the occurrence of the savanna biome coincide clearly across Africa, Australia, and South America (24).

Identification and Mapping of Key Fire Characteristics

Five important fire regime characteristics were identified and quantified globally at 0.5° resolution by using available remotely sensed data. These are fire return interval (FRI), maximum fire intensity, length of the fire season, maximum fire size, and mean annual area burned (Fig. S1 *A–E*), and they represent a significant advancement in the mining of satellite-derived information on fire—moving beyond indices of burned area to describe other ecologically meaningful metrics of fire (25) (*Materials and Methods*).

Fire intensity is a measure of the rate of energy released by fire, and FRIs are indicative of the growth period available to plants between fires. Changes in frequency or intensity of fires can alter vegetation community composition by promoting or excluding different types of plants (1, 2), and are important for feedbacks between vegetation and fire (26, 27). The size of individual fires and the length of the year over which fires occur reflect the continuity of the fuel bed and the propensity of the vegetation to be flammable (28, 29). Mean burned area is a widely used global measure of fire. It has been linked to the paleorecord (30), and is used to calculate fluxes of carbon from the biosphere to the atmosphere (3). Distinguishing fire type is important in assessing fire regimes, and we hoped that crown and ground fires would emerge as having distinctive combinations of fire characteristics.

Two of these datasets—fire size and FRI—have not been produced globally before. Previously published works (25, 31) and *SI Materials and Methods* detail their derivation and validation [the FRI metric is most reliable for return intervals less than 50 y (Fig. S2)].

Results and Discussion

Physical Limits of Fire. As expected, there was structure in multidimensional fire space and tradeoffs between particular fire characteristics. Fire intensity is strongly constrained where fires are frequent, but infrequent fires show a wide range of fire intensities (Fig. 1*A*). Ecologically, infrequent fire is associated with intense crown fire events and with low-intensity surface fires in tropical forests. When fire is frequent, there is insufficient time to accumulate the fuel needed for very high-intensity fires, and these systems are climatically too flammable to produce creeping, low-intensity fires.

Very long fire seasons are only associated with small fires (Fig. 1*B*). Long fire seasons are typical of human-derived management fires (28), which are also likely to be small (32), so this section of fire space might be a human creation.

Constraints between maximum fire size and area burned are not as tight as would be expected from studies that have found the majority of area to be burned in a few large fires (33). High burned areas can result from a range of fire size distributions (Fig. 1*C*). However, regions with very small fires do show slightly lower burned area. Finally, systems with short FRIs always show high area burned, but, at longer FRIs, the area burned ranges more widely (Fig. 1*D*). A complete set of bivariate relationships is shown in Fig. S3.

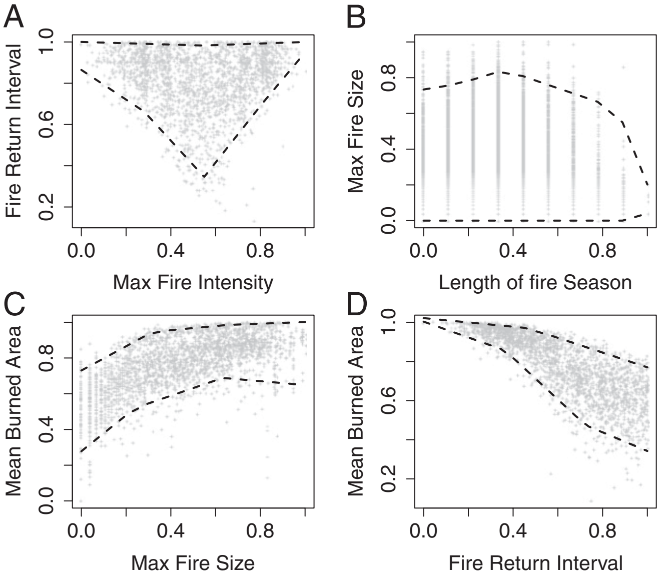

Fig. 1. Multidimensional fire space represented by selected combinations of fire characteristics. FRI by maximum fire intensity (*A*), fire season length by maximum fire size (*B*), maximum fire size by mean burned area (*C*), and FRI by mean burned area (*D*). For each combination there are constraints—presumably imposed through vegetation, climate, and people—which mean that not all of the space is occupied. Data are logged and rescaled. Dashed lines represent fifth and 95th piecewise quantile regression fits to the data. Fig. S3 shows graphs of all combinations.

These observed patterns challenge some of the prevailing assumptions around global fire relationships used to drive fire models and estimate fire emissions. Thonicke et al. (34) assume a correlation between fire season and area burned that is not apparent in our data, and Hao and Liu (35) overestimated emissions from Africa by assuming a log-linear relationship between FRI and area burned.

The results raise important questions about how amenable these patterns are to manipulation as a result of changes in climate or vegetation or by human management. It is possible that humans, by initiating burning of small agricultural fires at unpredictable times of year (28), pushed the system further into the lower right corner (i.e., small fires, long fire season) of Fig. 1*B* than it was before. Similarly, before C4 grasses evolved, very short annual and subannual FRIs (Fig. 1*A*) might not have existed on the globe.

Identifying Pyromes. We identified five distinct pyromes (*SI Materials and Methods*), each associated with particular combinations of fire characteristics (Fig. 2). Two of these pyromes have high annual burned areas and frequent fires (Table 1 and Fig. 3). These pyromes differ in their fire size and fire intensity, as well as their spatial distribution: Australia has larger, more intense fires [frequent–intense–large (FIL)], whereas in Africa, smaller, less intense fires dominate [frequent–cool–small (FCS)] (Fig. 2). Two more pyromes were identified with infrequent fires and very short fire seasons (Table 1 and Fig. 3). One has high-intensity, larger fires [rare–intense–large (RIL)], and the other has lower fire intensity and smaller fires [rare–cool–small (RCS)]. These pyromes dominate in temperate and boreal regions but occur elsewhere as well (Fig. 2).

A final pyrome was identified with intermediate fire return times but fairly small fires (Table 1 and Fig. 3). This intermediate–cool–small (ICS) pyrome occurs throughout the globe, particularly in regions of deforestation and agriculture (Fig. 2).

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Table 1. Characteristics of the five pyromes identified by model-based expectation–maximization clustering

Characteristic	FIL (yellow)	FCS (orange)	RIL (green)	RCS (purple)	ICS (blue)
Mean burned area, %	14 (8–36)	9 (3–17)	1 (0–2)	0 (0–0.5)	0 (0–1)
Estimated FRI, y	3 (1–4)	1 (1–2)	>50	>50	12 (6–19)
Max FRP, MW	473 (350–660)	197 (156–253)	476 (283–844)	187 (108–334)	224 (143–352)
Max fire size, km^2	414 (155–1437)	25 (15–43)	83 (38–214)	4 (2–9)	9 (5–17)
Length of fire season, mo	4 (3–4)	3 (3–4)	2 (1–2)	1 (0–1)	3 (3–4)
Tropical moist broadleaf forests	2	10	5	**28**	**56**
Tropical dry broadleaf forests	3	21	6	16	**53**
Tropical coniferous forests	0	10	13	17	**59**
Temperate mixed forests	1	2	11	**41**	**45**
Temperate coniferous forests	2	0	**26**	**45**	**26**
Boreal forests	1	0	**46**	**47**	6
Tropical grasslands and shrublands	**29**	**33**	9	6	**23**
Temperate grasslands and shrublands	10	3	21	**29**	**37**
Flooded grasslands	**46**	17	8	5	**24**
Montane grasslands	6	15	13	**31**	**36**
Mediterranean vegetation	1	3	**25**	**36**	**35**
Xeric vegetation	18	1	**31**	**24**	**26**

The median and 25th to 75th quantiles of each fire characteristic are reported (parentheses), as are the percentages of 12 World Wildlife Fund biome classes that fall into each pyrome (values greater than 20% are bold, Table S1 shows for pyromes by biome).

The largest fires are found in the FIL pyrome (414 km^2 on average; Table 1), and the most intense fires are found in the RIL pyrome (476 MW). In FCS regions fires occur annually, whereas the RCS pyrome has an average FRI of more than 100 y. The shortest fire seasons are associated with RIL and RCS pyromes (less than 2 mo), and average area burned is hardly detectable in the RCS and ICS pyromes.

Correspondence of Pyromes to Biomes. Chuvieco et al. (9), by using three active fire data metrics, successfully distinguished boreal forest but found no clear patterns with other vegetation classes. In our study, more than 93% of the boreal forest falls into the RIL and RCS pyromes (green and purple, respectively) and we also found the FIL and FCS (yellow and orange, respectively) pyromes to be concentrated in tropical grasslands (Figs. 2–5). In contrast, xeric vegetation is associated with four of the five pyrome classes (Table 1), presumably depending on the type of fuels present in the particular arid system (shrubs, grasses, or tussock grasses).

Fires in tropical moist broadleaf forests are predominantly of the ICS (blue) type but this biome can also sustain RCS (purple) and FCS (orange) fires (Table 1). Almost all biomes contain a substantial proportion (>20%) of the ICS pyrome (Table 1). Our analysis did not distinguish between North American and Russian boreal fire regimes, although the fire intensity data showed the North American fires to be generally more intense (Fig. S1*C*).

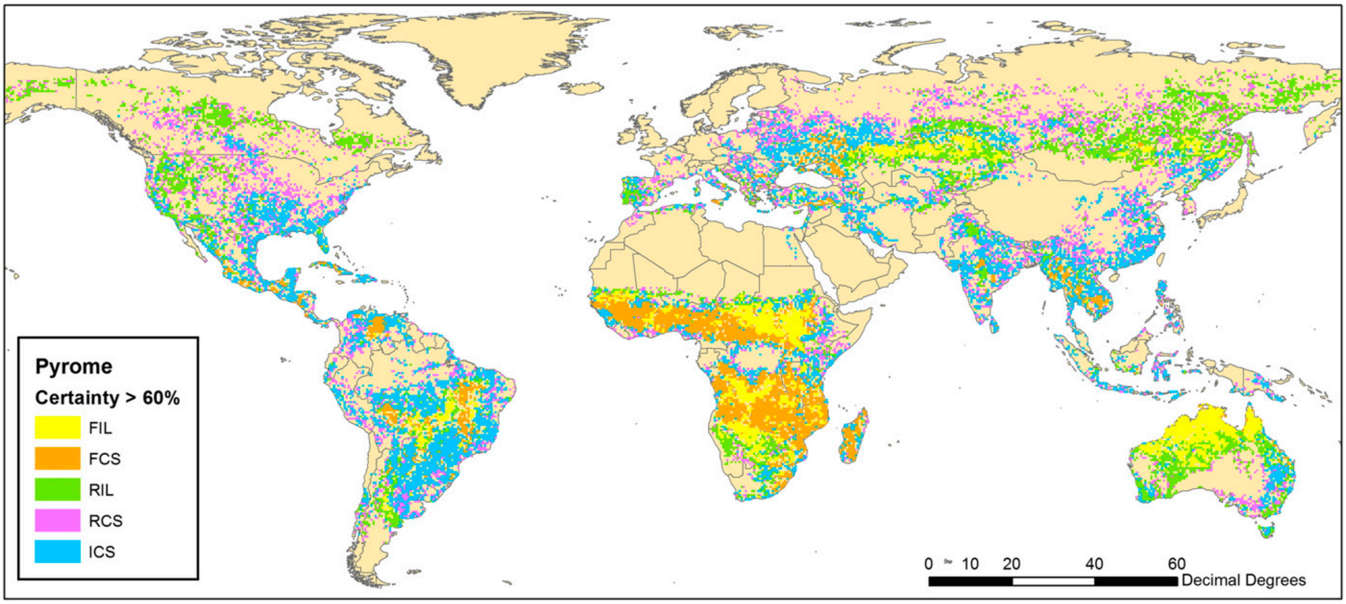

Fig. 2. Mapping the spatial distribution of pyromes. Produced from the five-cluster solution of a model-based expectation–maximization clustering algorithm. Pyromes represent regions of the globe that have similar fire frequencies, intensities, sizes, burned areas, and fire season lengths. Pixels with greater than 60% probability of being uniquely categorized are plotted (85% of the data).

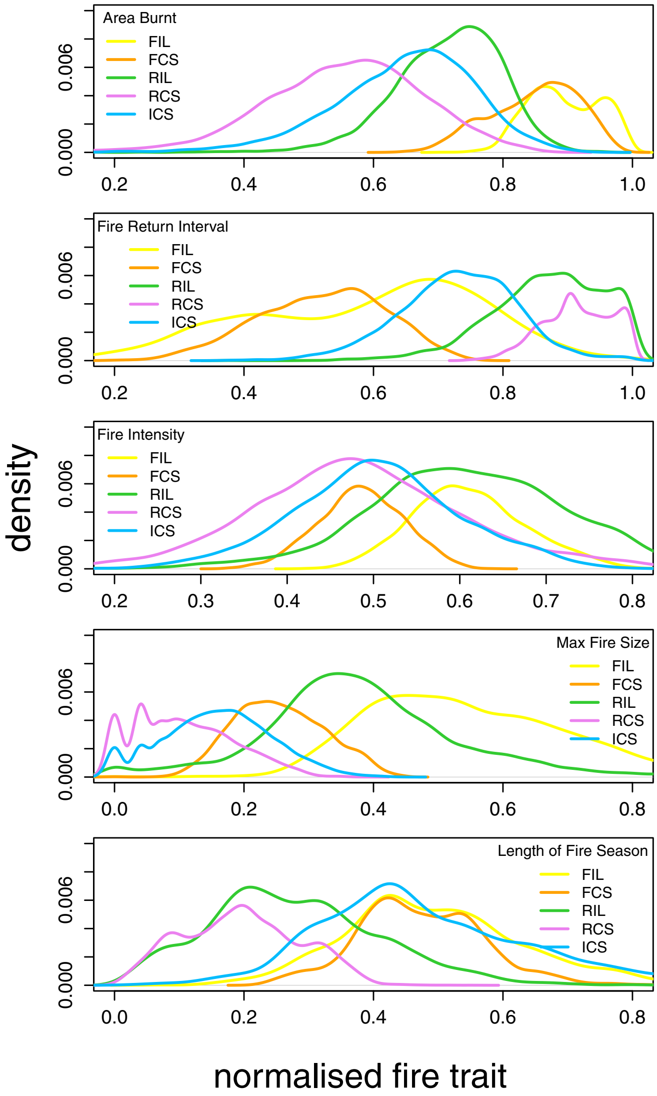

Fig. 3. Distinguishing pyromes using frequency distributions of fire characteristics. Data are logged and rescaled. Table 1 shows real values.

Environmental Correlates. Axes of mean annual temperature (MAT) and mean annual precipitation (MAP) have traditionally been used to describe the climatic determinants of biomes (18), but these metrics are not necessarily as strongly associated with pyromes. We identified three climate variables that would be expected to explain spatial variability in fire regimes: effective rainfall (i.e., productivity), interannual variability of rainfall (i.e., probability of long-term droughts affecting woody fuels), and seasonality of rainfall (i.e., probability of short-term droughts affecting grassy/litter fuels).

The five pyromes overlap substantially when plotted on axes of MAP and MAT (Fig. 4*A*), although it does appear that the two frequently burning pyromes (FIL and FCS) cover warmer, wetter climate space than the other pyromes. When plotted on axes that are more meaningful to fire, this inference is shown to be incorrect (Fig. 4*B*). Actually, frequently burning pyromes dominate in more arid ecosystems (measured by effective rainfall) than other pyromes. These pyromes are also associated with strong rainfall seasonality, which provides frequent (annual) opportunities for fire (Fig. 5).

Human impacts (HIs) on fire are globally apparent (30, 36). To examine this, we used an HI score that integrates information on human densities and land transformation (37). The HI score clearly distinguishes the ICS from the other pyromes. Crucially, we also found that pyromes characterized by large fires (FIL and RIL) are correlated with low HI (Fig. 5). These data suggest that human activities are a major force disrupting the fire system, and that human activities play an equal role to climate in determining current pyromes. For example, the difference between the two grassy fire regimes (FIL and FCS) is most clearly explained by their differences in HI than by any climatic drivers, and they probably represent a “sparsely habited grassy pyrome” and a “densely habited grassy pyrome,” respectively.

Relation of Pyromes to Known Fire Regimes. Despite the absence of clear correspondence among climate, biomes, and pyromes, a basic dichotomy between the pairs of frequently burnt vs. infrequently burnt pyromes can be distinguished (Fig. 3). This dichotomy probably reflects a division between ecosystems dominated by alternative fuel types (grasses vs. litter).

Only grasses have the physiological capacity to regrow lost biomass quickly enough to sustain the frequent (i.e., every 1–3 y) fire regimes characteristic of the FIL and FCS pyromes. This is corroborated by their dominance in regions of tropical and temperate grassland (Fig. 2), and their association with high rainfall seasonality—a climatic characteristic of C4 grassy ecosystems (38). Respectively, 91% and 68% of these two pyromes are classified as grassland vegetation of some sort.

Only the RIL pyrome has the combined high fireline intensity (close to 500 MW) and long fire return times (>30 y) that would be associated with crown fires. This pyrome prevails in broadleaf and needleleaf forests, is found wherever there is Mediterranean vegetation (Iberian peninsula, west coast of the United States, southwestern Australia), and also in the spinifex grasslands of Australia (Fig. 2). It is often interspersed geographically with the RCS pyrome—presumably, surface fires burning through litter fuels in the same vegetation.

Reported FRIs for crown fire regimes range from approximately 15 y in fynbos (39) to approximately 150 y in conifer forests (40). Our data were unable to distinguish these and only identified one crown fire pyrome. With better global FRI data, more subtle distinctions in crown fires might become apparent. It appears, however, that the distinction between different types of crown fires is small relative to global variation in fire characteristics, and that there is convergence in the fuel characteristics, if not the vegetation characteristics of these divergent ecosystems.

The prevalence and ubiquity of the ICS pyrome suggest that this is a human-derived pyrome (41). It is characterized by very small fires (ranging from 5 to 17 km^2), which are a known consequence of human landscape transformation that creates physical barriers to fire spread (25, 32). In other characteristics, this pyrome represents a homogenization of the extreme characteristics of other pyromes (Fig. 3), and it cuts across vegetation types from tropical broadleafed forests to xeric vegetation (Table 1). This would be expected of a fire regime consisting of fires lit by people for specific purposes, and therefore less closely connected to vegetation and climate drivers.

Implications. These data and several recent papers (42, 43) support our contention that fire is unlikely to be unilaterally responsive to climate in a deterministic way. The pyromes characteristic of the modern world can be explained with reference to current vegetation and climate, but complex interactions among fire, climate, vegetation, and modern human activities impede our ability to predict emergent fire regimes.

Whitlock et al. (6), in recognizing these vegetation–fire–climate contingencies, suggest that a fire regime be defined as “the full range of variability in fire activity within a given vegetation type.” Our analysis shows that pyromes are not simply a construct of biomes. One biome can contain different fire regimes, and con-

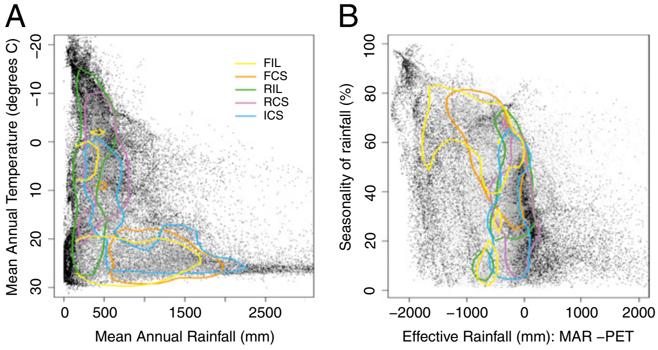

Fig. 4. Plotting pyromes in climate space. A Whittaker plot (MAP–MAT) does not clearly distinguish pyromes as it does with vegetation (*A*). Meaningful climate indices improve the separation (*B*), but pyromes are not determined by climate alone. Black points show all vegetated 0.5° grid cells, gray points show all cells that had fire data. Lines show the 95th quantile of the density of points for each pyrome class.

trasting biomes, such as xeric shrublands and boreal forests, may effectively converge on a common pyrome through similarity in the underlying dominant fuel type and growth constraints on fuel development.

There do appear to be predictable relationships between fuel types and fire characteristics, so developing spatial data on fuel characteristics (e.g., flammability of fuels, litter structure, propensity for crown fire) might help to clarify the links between vegetation and fire. Similarly, choosing meaningful climate indices that relate directly to how climate controls the amount and type of fuel and its propensity to be flammable would undoubtedly make climate–fire relationships clearer.

This spatial analysis corroborates meta-analyses of charcoal records, which suggest that HIs have acted to decouple the fire–climate signal in recent years (44). The current spatial extent of the human-modified pyrome (Fig. 2), and the superior predictive power of our HI score over climate indices (Fig. 5), point to the significant role humans can play in determining fire characteristics and driving future fire regimes.

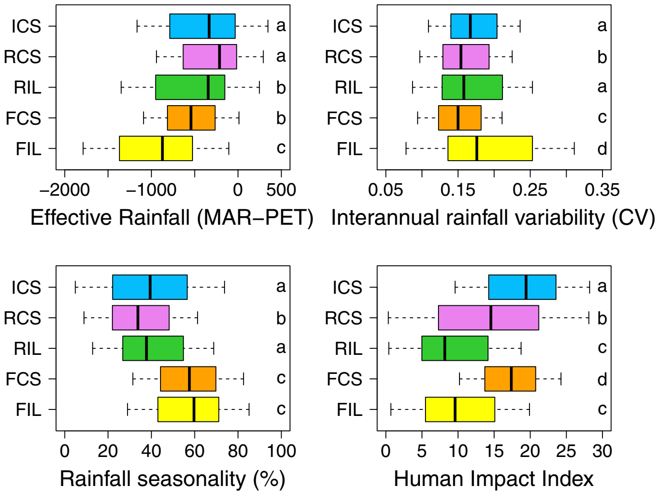

Fig. 5. Environmental characteristics of the five pyromes. The climate variables were chosen to represent important drivers of fire (see text and *SI Materials and Methods*). Lines represent the median, boxes the 25th and 75th quantiles, and whiskers the 0.5× interquartile range of the data. Significantly different distributions (two-sided *t* test) are indicated with letters.

The pyromes described here depend on the fire characteristics included, on the number of clusters chosen, and on the limitations of data used to represent each fire characteristic (*SI Materials and Methods*), and are unlikely to be definitive. Moreover, the 14-y data record effectively sets the time period over which we are describing global fire regimes, and is only a snapshot of a continuously changing system. However, the spatially explicit nature of our data and the range of fire metrics used complements longer-term, less data-rich studies (see ref. 45 for justification). Our pyromes could be useful to global fire modelers as a test of their ability to replicate the different types of fires apparent on the globe today. Moreover, the use of pyromes as well as biomes as categories to estimate emissions from wildfires could constrain some of the uncertainty—for example, by providing meaningful fire units for Intergovernmental Panel on Climate Change (IPCC) look-up tables.

Ultimately, we require confidence in our ability to infer past pyromes, and to predict patterns of world fire we might experience in the future. It is clear from this analysis that not all combinations of fire characteristics are possible (Fig. 1), and that various factors constrain modern fire regimes. These constraints appear to be defined by energetics: interactions between fuel type, climate, atmospheric chemistry, and rates of regrowth after fire (13). It is also clear that HIs, novel climates, and evolving vegetation can alter the type of fire that occurs in a location (15, 17, 27, 46–48). Whether these impacts are shifting the occurrence of fire within the multidimensional fire space identified here (Fig. 1), or releasing some of the important constraints on fire and allowing fire regimes to move into novel parts of this space is not yet clear.

Materials and Methods

Data Preparation. The datasets were produced using global remotely sensed estimates of active fire and burned area [moderate-resolution imaging spectroradiometer (MODIS) MCD45A1, MCS14ML, and Global Fire Emissions Database (GFED) 3.1]. These are available for a minimum of 10 y (operational MODIS sensors; however, the GFED3 burned area product provides a 14-y time series) and a minimum of 500 m spatial resolution. The short observational time period is an obvious constraint to characterizing fire regimes in systems which burn infrequently, but we chose indices that maximized the information that could be gained from these global datasets (*SI Materials and Methods*). The analysis was run for all 0.5° grid cells in which there was fire information (26,455 data points).

We summed the 0.5° GFED3.1 monthly burned area data to produce an annual measure of area burned from 1997 to 2010 and averaged this produce mean area burned (Fig. S1*A*). Satellite middle-IR wavelength measurements sensed over actively burning fires can quantify the rate of radiant energy release: the fire radiative power (FRP) (49). We used the MCD14ML active fire product to calculate the 95th quantile of FRP as a proxy for the maximum fireline intensity (Fig. S1*B*). FRI is usually estimated by fitting a Weibull distribution to fire return time data (50). This method resulted in convergence of FRI estimates for only some portions of our global FRI dataset (*SI Materials and Methods*). The coefficient of variation (mean/SD) in annual burned area (CVBA) correlates well with FRI ($R^2 = 0.67$; Fig. S2) because interannual variability in area burned is higher when fires occur infrequently. We calculated CVBA from the GFED3.1 data product (Fig. S1*C*). Individual fires can be identified from burned area data by using a flood-fill algorithm (31). We did this for the globe by using the MCD45A1 burned area product (*SI Materials and Methods*) and used the 95th quantile of fire size in each grid cell to give maximum fire size (Fig. S1*D*). The length of the fire season was quantified as the number of months required to reach 80% of the total average annual burned area (Fig. S1*E*).

Climate characteristics were quantified from the 0.5° Climatic Research Unit (University of East Anglia) long-term global climate data (www.cru.uea.ac.uk/; accessed November 2011) and the WorldClim dataset (www.worldclim.org; accessed November 2011). The difference between MAP and mean annual potential evapotranspiration (i.e., effective rainfall) produces an index of productivity whereby anything less than zero indicates that evaporative demand exceeds incoming precipitation. We used a rainfall concentration index (51) to describe rainfall seasonality. Longer-term wetting and drying events can also impact the probability of fire (47), and to represent this we used the coefficient of variation in annual rainfall. We used the HI index (37) to represent the effects of people on ignitions and land connectivity. We used the WWF terrestrial ecoregions map, which is the only global product derived from ground-truthed vegetation maps, and which identifies 14 different global biomes. Although it uses climate qualifiers to classify biomes, which is something we were aiming to avoid in our pyromes classification, it gives a better description of the fuels than other modeled or remotely sensed biome products (*SI Materials and Methods*).

Analysis. Except where indicated, the nonnormal data (FRI, maximum FRP, CVBA, maximum fire size, mean burned area) were logged, and all data were rescaled between 0 and 1 and centered. A principal components analysis was run using the princomp package in the open-source R statistical software. All five fire characteristics contributed significantly to the principal components and were largely orthogonal to each other (Fig. S4). Constraints and tradeoffs between different fire characteristics were demonstrated by plotting each characteristic against the others, by using a piecewise quantile linear regression (and the R statistical package quantreg) to identify the fifth and 95th quantiles of the data. The Mclust package was used to identify regions of similar fire regimes (pyromes). This package allows statistical comparison of different cluster sizes and the shapes of the clusters using the Bayesian information criterion (BIC). Importantly, it also allows for estimation of uncertainty around classifications by using expectation maximization methods (52), and for predicting the grouping of points not used in the original analysis based on their attributes. We used a default prior with two clusters and a variable shape, variable volume, variable orientation (VVV) model to regularize the fit to the data. An initial analysis using all possible parameterizations of the covariance matrix and cluster sizes from two to 15 indicated that a VVV model gave the best results (i.e., highest BIC). Choosing the correct number of clusters (i.e., pyromes) was difficult because the BIC continued to increase to 15 clusters. Beyond five clusters, the increase in BIC was negligible (Fig. S5) and did not justify further splitting of the data. As a result of processor constraints, we ran Mclust on a training sample of 10,900 points and predicted the pyrome of the remaining approximately 15,000 points by using discriminant analysis (mclustDAtrain and mclustDAtest). All points were classified into one of five pyromes with an associated uncertainty value. Uncertainties were fairly constant across pyromes, with most points having uncertainty values less than 0.2 and no clear spatial pattern emerging (Fig. S6). Fig. S7 indicates how the five pyromes emerged from the clustering procedure.

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